TRYOLABS IT ADMINISTRATOR PROVISIONING SPECIFICATION • ZERO-PRIVILEGE ATTENDEE MODE

Delivered by Tryolabs • Powered by Google Antigravity & Gemini Enterprise

IT Administrator Workstation Requirements & Pre-Provisioning Guide

Complete OS, CLI Binary, Python Library (requirements.txt), Port, EDR & File Permission Specification for Non-Admin Workshop Computers (Telecom & Retail Editions)

CRITICAL REQUIREMENT: ATTENDEES DO NOT HAVE INSTALL PERMISSIONS
Workshop attendees will use locked-down corporate computers without local administrator (**sudo** / Administrator) privileges. IT Administrators must pre-install the system binaries below and pre-build the repository **.venv** virtual environment using **admin_provision_workstation.sh** (macOS/Linux) or **admin_provision_workstation.ps1** (Windows), then grant recursive read/write ownership of the workshop repository directory to the standard attendee user account.

1. Hardware, OS, Local Loopback Ports & EDR/Firewall Rules

Category	Requirement Specification
Operating System	macOS 13+ (Ventura/Sonoma/Sequoia), Windows 10/11 (64-bit) with Git Bash or WSL2, or Linux (Ubuntu 22.04+, Debian 12+, RHEL 9+)
CPU & Memory (RAM)	Minimum 4 vCPU / 8 GB RAM (8+ vCPU / 16 GB RAM recommended); 10 GB free disk space
Local TCP Ports (Loopback Only)	127.0.0.1:8001 (Telecom UI), 127.0.0.1:8002 (Retail UI), and 127.0.0.1:8080 (Google ADK Dev UI **adk web**) — Unprivileged local binding only
EDR / Antivirus Subprocess Rule	Allow **.venv/bin/python** to run as a child process over **stdio** inside the workshop repository for FastMCP (**mcp_server/**) and **PreToolUse** hooks (**scripts/audit_hook.py**)
Outbound HTTPS Allowlist (443)	***.googleapis.com** (**cloudaicompanion**, **aiplatform**, **discoveryengine**, **run**), **accounts.google.com**, **oauth2.googleapis.com**

2. Complete System Software & CLI Binary Matrix (Must Be in System PATH)

Software / Binary	Version	Verify Command	Workshop Role	Status
1. Google Antigravity	Latest Enterprise	`agy --version` / App Launch	Antigravity 2.0 Desktop App, Antigravity IDE, and `agy` CLI in PATH	REQUIRED
2. Python 3 Runtime	>= 3.11 (>= 3.10 ok)	`python3 --version`	Must include standard modules `venv`, `pip`, `sqlite3`, `decimal`	REQUIRED
3. Astral `uv`	>= 0.5.0	`uv --version`	Fast Python environment & package manager in system PATH	REQUIRED
4. Git CLI	>= 2.39	`git --version`	Repository cloning, branching, and diff inspection	REQUIRED
5. SQLite 3 CLI	>= 3.37	`sqlite3 --version`	Lab 1 & Lab 3 database inspection (`data/*.db`) and hook tests	REQUIRED
6. Bash & cURL	bash >= 3.2, curl	`bash --version`	Executes `./install.sh`, `./verify.sh`, and local A2A checks	REQUIRED
7. Terraform CLI	>= 1.5.0	`terraform -version`	Lab 4 Cloud Run & IAM validation (`infra/terraform/`)	REQUIRED (Lab 4)
8. Google Cloud SDK	>= 460.0.0	`gcloud --version`	Gemini Enterprise auth & Lab 4 deployment (`gcloud auth`)	REQUIRED (Lab 4)
9. Chromium Browser	Chrome / Edge	Open browser	Viewing `:8001`/`:8002` dashboard, `:8080 adk web`, & HTML slides	REQUIRED

3. Complete Python Library Specification (`requirements.txt`)

All Python libraries below must be pre-installed by the IT Administrator inside the repository's `.venv` virtual environment so that non-admin attendees never need to run `pip install` during the workshop:

Python Package	Min Version	Locked Version	Usage Across Workshop Labs 1–4
`fastapi`	>=0.110.0	0.141.1	Core REST API & A2A JSON-RPC 2.0 endpoint (`/api/v1/a2a`) across Labs 1–4
`uvicorn`	>=0.27.0	0.54.0	Local ASGI server for live Telecom NOC (`:8001`) and Retail Pulse (`:8002`) dashboards
`pydantic`	>=2.6.0	2.14.1	Strict data & schema validation for CDRs, 5G slices, WMS inventory, and A2A payloads
`pytest`	>=8.0.0	9.1.1	Automated verification suite executed by `./verify.sh` across Labs 1, 2, 3, and 4
`httpx`	>=0.27.0	0.28.1	FastAPI `TestClient` transport and async A2A protocol requests
`mcp`	>=1.2.0	2.3.0	Official Model Context Protocol SDK (`FastMCP`) for Lab 3 local `stdio` MCP server

Python Package	Min Version	Locked Version	Usage Across Workshop Labs 1–4
`google-adk`	>=1.0.0	2.12.0	Google Agent Development Kit (**`LlmAgent`**, **`Runner`**, and **`adk web`** Dev UI on **`:8080`**)
`google-genai`	>=1.0.0	2.33.0	Google GenAI / Vertex AI client SDK used by ADK multi-agent hierarchies
`google-auth`	>=2.28.0	2.62.0	Google Cloud IAM & Application Default Credentials authentication
`aiosqlite`	>=0.20.0	0.22.1	Async SQLite session storage backend required by **`adk web`**
`sse-starlette` & **`starlette`**	>=2.0.0 / >=0.36.0	3.5.0 / 1.9.0	ASGI routing and Server-Sent Events streaming for FastMCP and ADK
`websockets`	>=12.0	15.0.1	Real-time bidirectional event streaming in **`adk web`**
`python-dotenv` & **`PyYAML`**	>=1.0.0 / >=6.0.1	1.2.4 / 6.0.3	Environment configuration and YAML frontmatter parsing for Lab 2 **`SKILL.md`**
`python-multipart` & **`requests`**	>=0.0.9 / >=2.31.0	0.0.32 / 2.34.2	FastAPI form handling and HTTP utilities
`tenacity`, **`watchdog`**, **`graphviz`**	>=8.2 / >=4.0 / >=0.20	9.1.4 / 6.0.0 / 0.21	Agent retry resilience, file watching, and ADK agent graph rendering
`opentelemetry-api` & **`sdk`**	>=1.24.0	1.42.1	Multi-agent execution trace telemetry inspected in **`adk web`**

4. Exact `requirements.txt` & Automated IT Admin Provisioning Commands

Copy-paste or use the included **`requirements.txt`** (and **`requirements-lock.txt`** for air-gapped wheelhouses) located in **`admin_requirements/`** and the repository root:

```
# requirements.txt — Tryolabs Google Antigravity Workshop (Labs 01-04)
fastapi>=0.110.0
uvicorn>=0.27.0
starlette>=0.36.0
pydantic>=2.6.0
python-multipart>=0.0.9
python-dotenv>=1.0.0
pytest>=8.0.0
httpx>=0.27.0
anyio>=4.0.0
requests>=2.31.0
mcp>=1.2.0
sse-starlette>=2.0.0
google-adk>=1.0.0
google-genai>=1.0.0
google-auth>=2.28.0
aiosqlite>=0.20.0
websockets>=12.0
PyYAML>=6.0.1
tenacity>=8.2.0
watchdog>=4.0.0
graphviz>=0.20
opentelemetry-api>=1.24.0
opentelemetry-sdk>=1.24.0
```

5. Critical File Permissions for Non-Admin Attendees & One-Step Provisioning

Because attendees run under unprivileged OS user accounts, **the attendee's user account MUST have full read/write/modify permissions** on the entire workshop repository folder (**`.venv/`**, **`data/`**, **`.agents/`**, **`src/`**, **`mcp_server/`**, **`adk_agents/`**, **`labs/`**). Run our automated provisioning script as Admin to build **`.venv`**, seed SQLite, and set user ownership automatically:

```
# Option A: Automated IT Admin Provisioning & Ownership Assignment (macOS / Linux)
sudo ./admin_requirements/admin_provision_workstation.sh --user <attendee_username>

# Option B: Automated IT Admin Provisioning & ACL Assignment (Windows PowerShell as Admin)
.\admin_requirements\admin_provision_workstation.ps1 -AttendeeUser "CORP\<attendee_username>"

# Option C: Verify Readiness as the Unprivileged Attendee User (Zero-Install Mode)
./admin_requirements/admin_provision_workstation.sh --verify-only
```

6. Pre-Delivery IT Admin Verification Checklist

#	Check	Pass Criteria (Run as Standard Non-Admin Attendee User)
1	System Binaries in PATH	**`python3`**, **`uv`**, **`git`**, **`sqlite3`**, **`bash`**, **`curl`**, **`terraform`**, **`gcloud`**, and **`agy`** resolve in standard user PATH
2	Pre-Built **`.venv`** Ready	**`./.venv/bin/python -c 'import fastapi, uvicorn, pydantic, pytest, httpx, mcp, google.adk, google.genai'`** exits 0
3	ADK Dev UI Binary Ready	**`./.venv/bin/adk --version`** and **`./scripts/run_adk_web.sh --check`** exit 0
4	Zero-Install Mode Active	Running **`./install.sh`** as standard user skips network downloads (**`[OK] Pre-provisioned .venv detected`**) and seeds **`data/*.db`**
5	Non-Admin Write Access	Standard attendee user can write to **`data/`**, **`.agents/`**, **`src/`**, and **`mcp_server/`** without **`Permission denied`** errors